Safety Roads.

2026

Final Year Project Presentation

AI-Based Pothole Detection & Smart Navigation System

Presented by

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ultralytics YOLOv8

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The Problem We Solve

Road conditions in many areas are poorly maintained, leading to frequent potholes that cause accidents, vehicle damage, and traffic disruptions.
Drivers lack real-time information about road hazards, and existing reporting systems are slow, manual, and inefficient.

- High accident risk
- Vehicle damage & maintenance cost
- No real-time alerts
- Inefficient manual monitoring

Key Challenges in Existing Road Systems

01.

No Real-Time Detection

Current systems fail to detect potholes instantly, leading to delayed identification and increased risk for drivers.
→ Delayed detection directly increases accident probability and road damage impact.

02.

Inefficient Reporting Systems

Manual inspections and reporting are slow, labor-intensive, and not scalable for large road networks.
→ Lack of automation results in poor maintenance response and delayed repairs.

03.

No Navigation-Based Alerts

Drivers are not warned about potholes in advance, causing accidents, vehicle damage, and unsafe travel.
→ Absence of proactive alerts prevents drivers from taking preventive actions.

Proposed Solution

AI-Based Detection System

Real-time pothole detection using YOLOv8 model integrated with a mobile application for instant analysis.

Smart Backend Processing

FastAPI-based backend processes images, applies filters, and returns accurate detection results efficiently.

Intelligent Navigation & Alerts

Map-based visualization with GPS tracking and voice alerts to warn users about potholes in real time.

Introducing the Safety Roads

Safety Roads is an AI-powered mobile application that detects potholes in real time using YOLOv8 and provides smart navigation with GPS tracking and voice alerts. It ensures safer travel by informing users about road hazards instantly.

Key Features of the System

"Our system provides real-time pothole detection, map visualization, and voice alerts. It also integrates GPS tracking, applies smart filtering to improve accuracy, and supports offline storage for better usability."

Real-Time Pothole Detection

Captures images using camera/gallery and detects potholes instantly using YOLOv8.

Interactive Map Visualization

Displays detected potholes on map with GPS coordinates and severity-based color markers.

Voice Alert System

Provides real-time voice warnings when approaching potholes during navigation.

GPS Integration

Automatically captures and stores location data for each detected pothole.

Smart Filtering System

Applies confidence, area, and aspect ratio filters to reduce false detections.

Offline Data Storage

Stores detected potholes locally using AsyncStorage for future access.

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System Architecture & Data Flow

FastAPI

- Camera Tab
- Map Tab
- Navigation Module
- Detection Service
- AsyncStorage

MOBILE APP (Expo)

- GET /health
- POST /detect
- YOLOv8 Detection Model
- Post-Processing Filters
- pothole_yolov83/best.pt

BACKEND (FastAPI)

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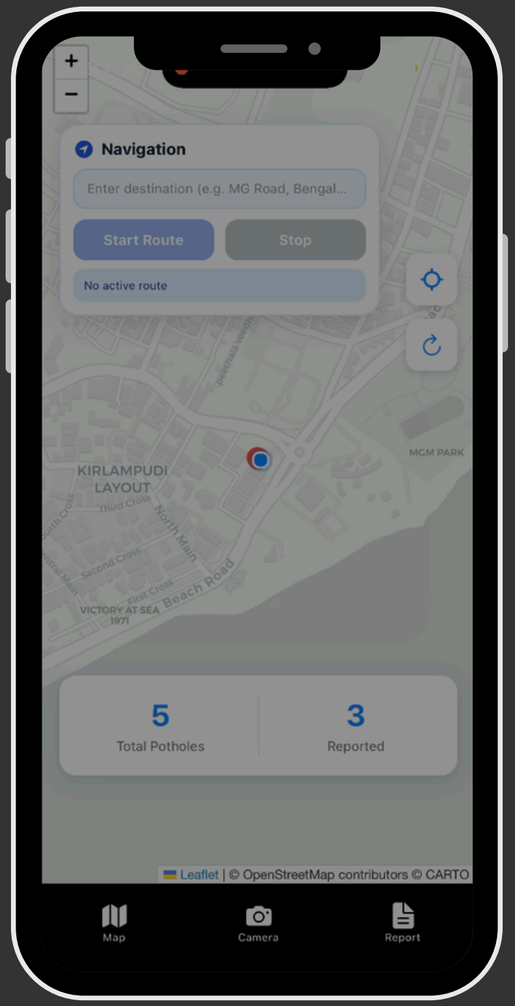
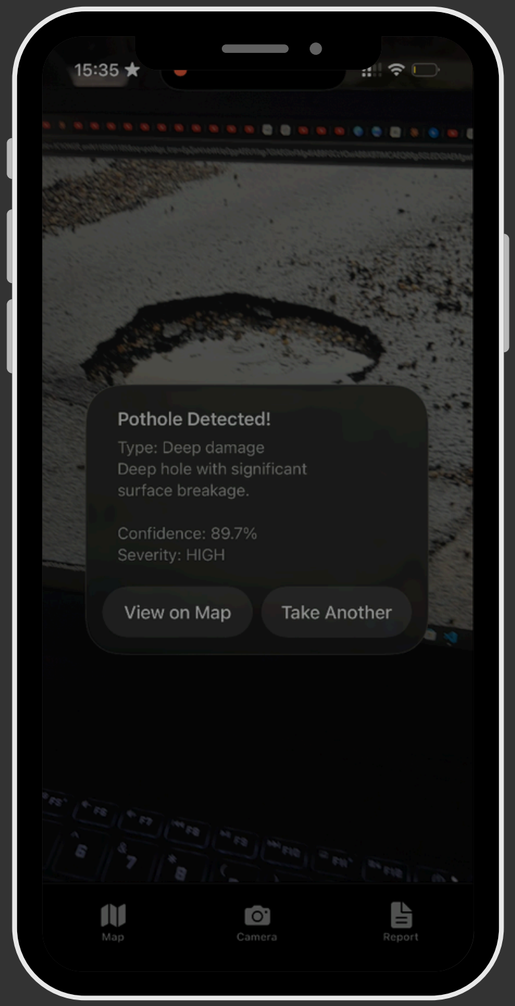
Mobile Application

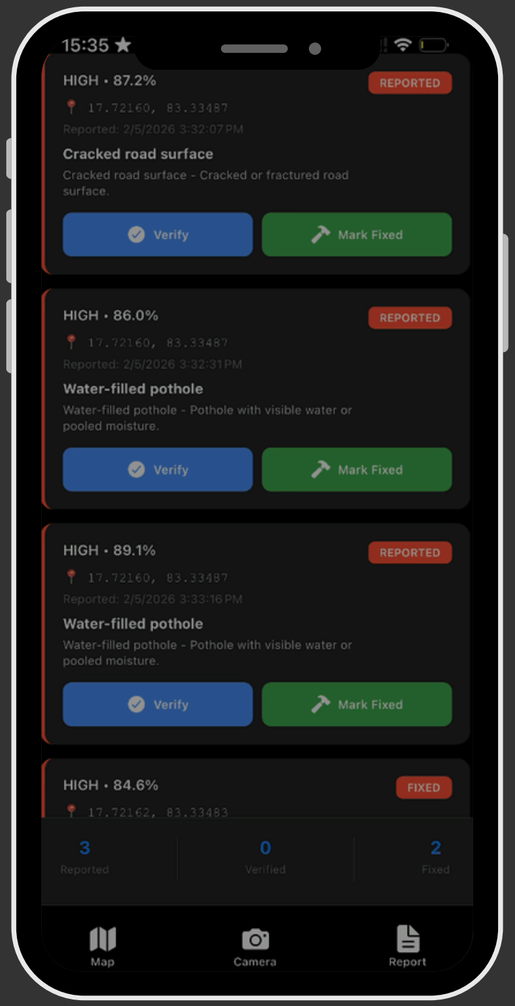
Maps



Camera



Reports



Backend System Overview

PHASE 1

- API Design
  - REST API using FastAPI
  - Endpoints: /health, /detect

PHASE 2

- Image Processing
  - Base64 decoding
  - PIL → NumPy conversion

PHASE 3

- AI Inference
  - YOLOv8 model detection
  - Bounding boxes + confidence

PHASE 4

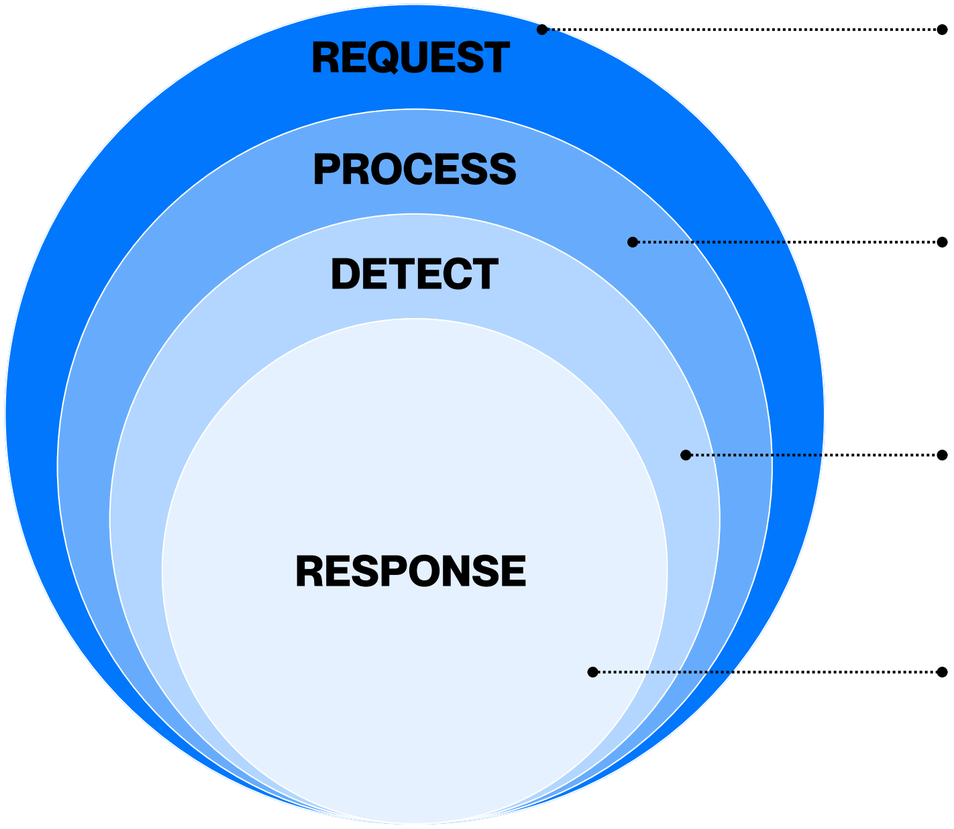
- Filtering Logic
  - Confidence threshold ≥ 0.65
  - Area & aspect ratio filtering

PHASE 5

- Response Generation
  - JSON output
  - Sent back to mobile app

API Workflow & Request Handling

Real-time API communication ensures fast and efficient pothole detection.



Request Handling

Mobile app sends image + GPS data as Base64 via POST request.

API Endpoints

- GET /health → Check model status
- POST /detect → Perform pothole detection

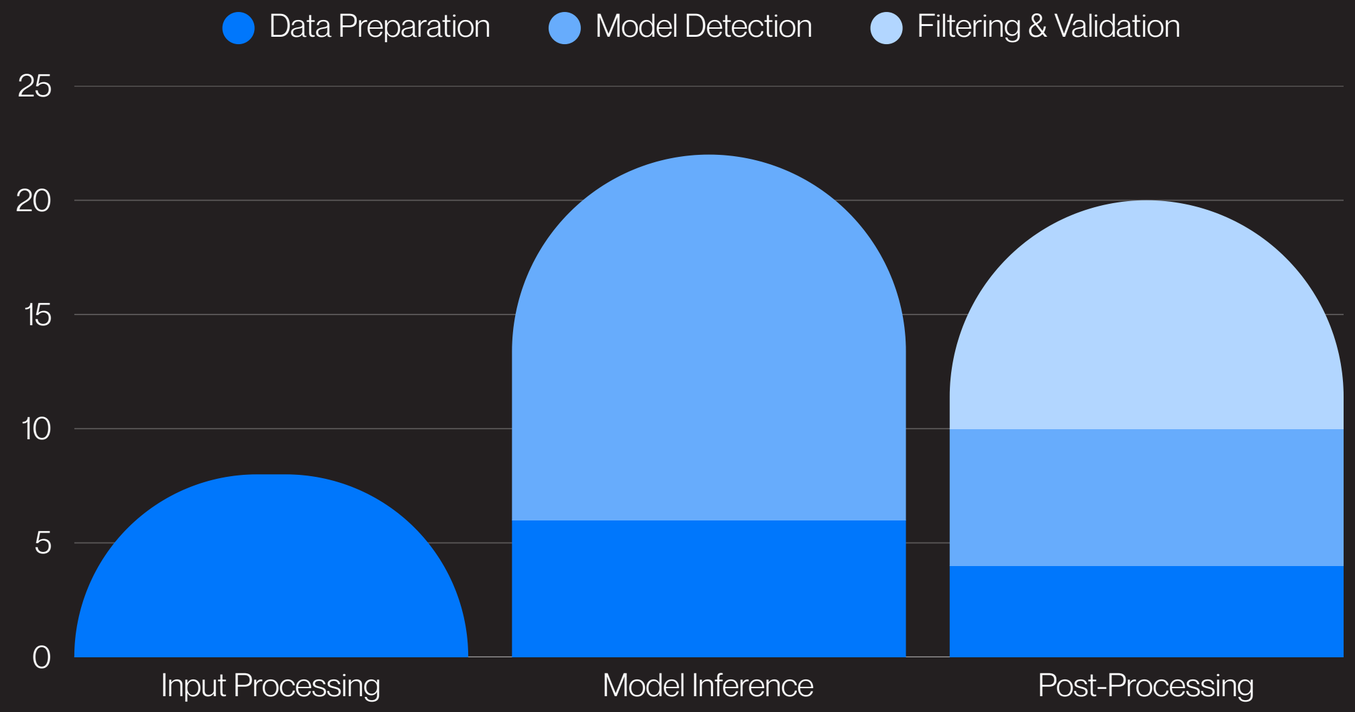
Backend Processing

Image is decoded, converted to NumPy, and passed to YOLOv8 model.

Response Generation

Detection results returned as JSON with confidence and bounding box.

YOLOv8 Inference Pipeline



Each stage of the pipeline contributes differently, with model inference being the most computationally intensive step.

The backend processes images through multiple stages including preprocessing, YOLOv8 inference, and post-processing. This pipeline ensures accurate and efficient pothole detection in real time.

Model Inference

YOLOv8 Nano processes images and predicts bounding boxes with confidence scores.

Detection Output

Extracts pothole locations, sizes, and classification results.

Filtering Stage

Applies confidence, area, and aspect ratio filters to remove false detections.

Post-Processing & Accuracy Optimization

Confidence Filtering

Removes detections below threshold (≥ 0.65) to ensure reliable predictions.

Dual-Layer Validation

Applies both server-side and client-side checks to improve robustness.

Bounding Box Area Filter

Rejects detections that are too small or too large (1% – 40% of image).

False Positive Reduction

Significantly reduces incorrect detections during real-world usage.

Aspect Ratio Constraint

Filters out irregular shapes that do not match pothole patterns.

Accuracy Improvement

Boosts effective precision from 61% → 75–80% after filtering.

“Our system applies multiple filtering techniques after detection to eliminate false positives and improve overall accuracy. This ensures reliable pothole detection in real-world conditions.”

Detection Data Pipeline — Step by Step

Image Capture

Camera / Gallery
GPS auto-captured
Base64 encoding

Health Check

GET /health
Model loaded?
Threshold config

Backend Detect

POST /detect
YOLO inference
Conf ≥ 0.65 filter

Client Validation

Conf ≥ 0.55
BBox check
Area ratio 1-40%

Storage

AsyncStorage save
GPS + severity
Timestamp record

Map + Alert

Marker on map
Color by severity
Voice proximity alert

🔴 High (>80%)

🟠 Medium (65-80%)

🟡 Low (<65%)

Dataset Information

"We used a dataset of 329 images with 284 annotations, split into training and validation sets. The dataset includes different types of potholes, which helps the model generalize well to real-world scenarios."

YOLO Annotation Format

```
# train2.txt
0  0.389  0.412  0.325  0.298
0  0.651  0.521  0.284  0.287
```

class_id x_center y_center width height

All coordinates normalized to [0, 1] range relative to image dimensions

80% Train / 20% Validation | Inter-annotator agreement > 95%

329
Total Images

264/65
Train / Val Split

284
Total Annotations

1
Class: Pothole

Future Enhancements & Conclusion

- Multi-Class Detection
- Federated Learning
- Municipal DB Integration
- Predictive Maintenance
- Cloud Crowd-Sourcing
- Night Vision Support

Project Achievements

- Custom YOLOv8 Nano — 73.22% accuracy, 25-50ms inference
- GPS-tagged interactive map with severity-coded markers
- Offline persistence with AsyncStorage — works without internet
- Full-stack Expo (React Native) + FastAPI production architecture
- Voice-guided navigation with pothole proximity alerts
- Production precision boosted to 75-80% via multi-stage filtering

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THANK YOU

AI-BASED POTHOLE DETECTION & NAVIGATION SYSTEM

**Delivering safer roads through intelligent,
real-time detection and navigation.**